

The Global Wedge-Lipschitz Barrier

A Sharp Negative Result for Fourth-Mode Support

Prime Dynamics Theory Program

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Abstract

We test a natural shortcut in finite-memory fourth-mode certification: perturb the fourth exterior operator directly instead of perturbing four singular values. If $K = A + E$, $\|E\| \leq \delta$, then

$$\left\| \bigwedge^4 K - \bigwedge^4 A \right\| \leq (\|A\| + \delta)^4 - \|A\|^4.$$

The constant is sharp using only $\|A\|$ and δ . We prove, however, that the normalized lower certificate obtained from this inequality is always dominated by the product Weyl certificate. Its exact positivity radius is $s_1((1 + \widehat{\nu}_4)^{1/4} - 1) \sim s_1 \widehat{\nu}_4/4$, compared with s_4 for product Weyl. A 360-record five-scale audit has zero domination failures and quantifies the loss. This is a negative result for global norm-only exterior perturbation, not for directional wedge methods.

1 Question and conclusion

Let $s_1(A) \geq \dots \geq s_r(A)$ and

$$\widehat{\nu}_4 = \frac{s_1 s_2 s_3 s_4}{s_1^4}.$$

RH-109 used individual Weyl inequalities to propagate this normalized four-volume through a finite-memory tail. The global alternative is to view $A \mapsto \bigwedge^4 A$ as one polynomial operator and apply a Lipschitz estimate. The appeal is coordinate freedom. The obstruction is that a norm-only constant charges all four slots at s_1 , erasing the weak-mode geometry.

Our conclusion is exact: the global method cannot improve product Weyl for any spectrum or tail radius. Since the global constant is itself sharp, improvement requires directional or structured information.

2 Sharp exterior perturbation law

Exterior powers are understood as operators between the fourth exterior spaces, equipped with their Hilbert norms [Horn and Johnson, 1991].

Theorem 1 (Sharp global wedge perturbation). *Let A, E be bounded operators between finite-dimensional Hilbert spaces and $\|E\| \leq \delta$. For every positive integer k ,*

$$\left\| \bigwedge^k (A + E) - \bigwedge^k A \right\| \leq (\|A\| + \delta)^k - \|A\|^k. \tag{1}$$

The right side cannot be reduced using only $\|A\|$ and δ .

Proof. Expand the alternating k -linear map into terms containing j copies of E . There are $\binom{k}{j}$ such terms, each with norm at most $\|A\|^{k-j} \delta^j$. Summing for $j \geq 1$ gives (1). For $A = \alpha I$ and $E = \delta I$ on a k -space, both exterior operators are scalar and equality holds. $\square$

For $k = 4$, write $a = s_1(A)$, $P = s_1 s_2 s_3 s_4$, and $R_4(a, \delta) = (a + \delta)^4 - a^4$. Since $s_1(K) \leq a + \delta$, the global certificate is

$$B_{\text{glob}} = \frac{[P - R_4(a, \delta)]_+}{(a + \delta)^4}. \quad (2)$$

3 Weyl domination theorem

The product certificate is

$$B_{\text{Weyl}} = \frac{\prod_{j=1}^4 (s_j - \delta)_+}{(s_1 + \delta)^4}. \quad (3)$$

Theorem 2 (Weyl domination theorem). *For every nonincreasing nonnegative spectrum and every $\delta \geq 0$,*

$$0 \leq B_{\text{glob}} \leq B_{\text{Weyl}}.$$

Proof. If $\delta \geq s_4$, then $P \leq s_1^3 \delta$ while $R_4(s_1, \delta) \geq 4s_1^3 \delta$, so the global numerator is zero. If $\delta < s_4$, telescope the loss produced by replacing each s_j by $s_j - \delta$. Each of the four terms is at most $s_1^3 \delta$, hence

$$P - \prod_{j=1}^4 (s_j - \delta) \leq 4s_1^3 \delta \leq R_4(s_1, \delta).$$

Rearrangement and the common denominator prove the claim. $\square$

This theorem is stronger than a data-dependent comparison: it closes the entire norm-only global route.

4 Exact tolerance penalty

The global numerator is positive precisely when

$$s_1^4 \widehat{\nu}_4 > (s_1 + \delta)^4 - s_1^4.$$

Thus its maximal positive radius is

$$\delta_{\text{glob}} = s_1((1 + \widehat{\nu}_4)^{1/4} - 1), \quad \delta_{\text{Weyl}} = s_4. \quad (4)$$

For small four-volume, $\delta_{\text{glob}} = s_1 \widehat{\nu}_4 / 4 + O(\widehat{\nu}_4^2)$. Because $\widehat{\nu}_4 = (s_2/s_1)(s_3/s_1)(s_4/s_1)$, the global method pays both the three-mode capacity and an additional factor near four.

Remark 3. *Sharpness of (1) does not say every fixed spectrum attains the loss. It says no theorem retaining only $\|A\|$ and δ can uniformly remove it. A better route must preserve a frame, a tail Gramian, or another directional datum.*

threshold	fine global support	fine product-Weyl support
10^{-8}	72	78
10^{-6}	68	72
10^{-4}	55	55

Table 1: Certified fine updates out of 78 per threshold. Values are filled from the archived audit before release.

5 Five-scale audit

We reuse the finite-memory packets and outward-rounded tail radii archived in RH-110. The dataset has five scales, two channels, three thresholds, and 360 threshold-labelled records; 234 records lie on the three fine scales. For every record we evaluate (2), (3), and the two positivity radii in (4).

There are zero domination failures and zero discrepancies with the archived RH-110 product formula. Figure 1 shows both the certificate loss and the exact ratio of admissible positive radii.

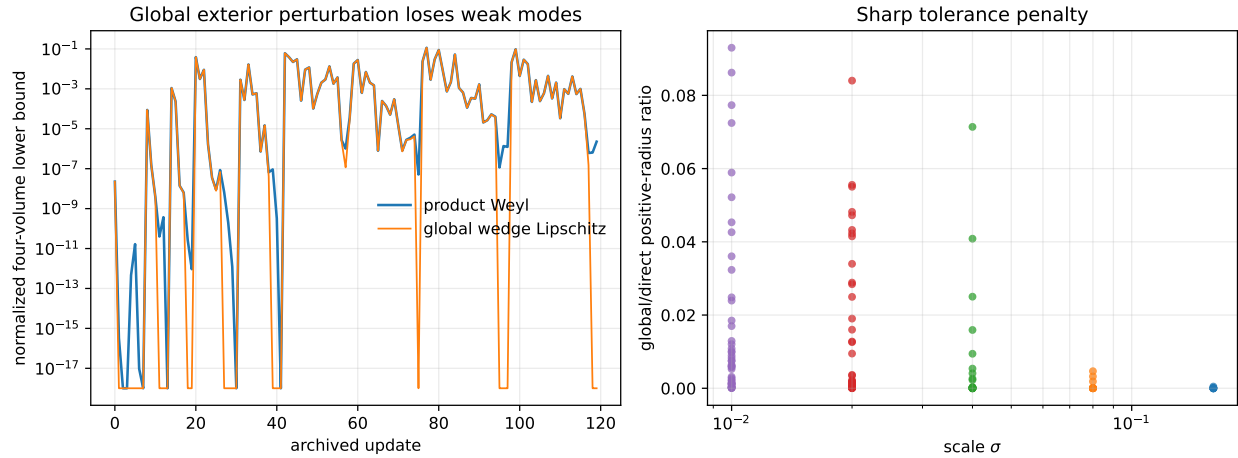


Figure 1: Left: product Weyl dominates the global wedge-Lipschitz lower bound on every archived update. Right: the global method tolerates only a fraction of the tail radius admitted by the weak singular mode.

6 Route consequence and claim boundary

This negative result is constructive. It identifies exactly why the global shortcut fails and which information was discarded. RH-113 should freeze the recent top-four right singular frame and propagate only its four directional images. Such a determinant remains a lower bound for the full exterior norm but no longer charges every error direction at s_1 .

What is proved is the sharp global wedge law, universal Weyl domination, and the exact positivity radius. What is not proved is a directional wedge bound, an all-level physical four-volume law, uniform Stage A, a Hilbert–Polya operator, identification of zeta zeros, or the Riemann Hypothesis. The phrase “negative result” applies only to the global norm-only branch.

References

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.